

PAL: A PRE-SPECIFIED PANEL OF ARTIFICIAL LISTENERS FOR PRONUNCIATION RECOVERABILITY MEASUREMENT

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ABSTRACT

Pronunciation assessment depends on expert-scored speech. PAL, a Panel of Artificial Listeners, measures how recoverable a known prompt is from a learner’s reading with heterogeneous recognizers: each recognizer’s phoneme error against the prompt is a measurement, and the equal-weight mean over 28 recognizers admitted by an a priori rule is the readout. No panel weight or score parameter is fitted to human ratings; ratings serve only development and validation. Design was fixed on the speechocean762 training split; five primary confirmatory comparisons were pre-specified and tested once after protocol commitment. PAL exceeds the best single recognizer on the held-out test split ($\Delta r = +0.038$ [$+0.021, +0.056$]), is segmental rather than prosodic on ERJ, replicates on EpaDB, correlates with human word recognition on ALLSSTAR ($r = 0.576$), and tracks Mandarin tone above consonant ratings. High-error recognizers retain most signal; none of the tested aggregation rules beats the mean. The scale is relative; grades need anchors.

Index Terms— pronunciation assessment, speech recognition, intelligibility, phoneme error rate, pre-specified protocol

1. INTRODUCTION

Supervised assessors such as GOPT and HMamba learn multiple rating aspects from expert-scored speech [1, 2], and every new learner population or language brings the same labeling cost [3]. We ask a measurement question instead of a prediction question: can the errors that a set of dissimilar recognizers make on a known prompt serve as a rating-free measurement of how recoverable a learner’s pronunciation is, and what does that measurement correspond to in human judgment?

Words are easier for some listeners to recover than for others [4], and intelligibility is a teaching target distinct from nativelikeness [5, 6]; recognizers that differ in architecture, training data, and decoding are probes with different failure modes. ASR-derived measures have long been associated with pronunciation ratings [7, 8], single-recognizer error features enter supervised assessors [9], ASR scores are combined

with acoustic predictors of accentedness and comprehensibility [10], and within-family Whisper ensembles feed supervised intelligibility predictors [11, 12]. A recognizer that recovers accented speech almost perfectly, however, has little variation left with which to rank pronunciations [13]. PAL is not an ensemble predictor but a pre-specified measurement instrument in which each recognizer’s errors are one repeated measurement of the same utterance: the errors themselves are the measurement, the panel is defined by an inclusion rule rather than by performance, and nothing is fitted to ratings.

The contribution is PAL, a Panel of Artificial Listeners, as a measurement instrument with a pre-specified validity study. PAL differs from recognition ensembles in objective and construction: it does not combine hypotheses to improve transcription, and it fits no predictor to ratings; recognizer-specific errors are repeated measurements of prompt recoverability, and the instrument is validated across learner populations, languages, and a direct human intelligibility criterion. Section 2 fixes the readout, the panel rule, seven hypotheses, and an unlock order for five corpora before any confirmatory data are read. Code, inventories, and the analysis plan are public.¹

2. MEASUREMENT DESIGN

2.1. Readout

For utterance u with prompt t , listener k returns a hypothesis $h_k(u)$. Prompt and hypothesis pass through the same normalization (lowercasing, ASCII punctuation other than apostrophes to spaces, digits to spoken names) and the same g2p_en conversion to stress-stripped ARPAbet. The phoneme error rate $e_k(u) \in [0, 1]$ is the Levenshtein distance divided by the reference length, capped at one; an empty hypothesis scores one. This transcript-derived PER compares canonical phone sequences of the recognized and prompted words, so it is a phone-weighted word distance, not a phone-level transcription of the learner’s realization. Phone recognizers are compared directly with eSpeak reference phones, without inven-

¹<https://github.com/mnaoizy/pal-panel>

tory mapping. The readout is

$$p(u) = -\frac{1}{K} \sum_{k=1}^K e_k(u), \quad (1)$$

an equal-weight mean over the K admitted listeners; larger $p(u)$ means greater recoverability. For Mandarin, characters are simplified and pinyin conversion yields separate syllable-base and tone sequences, giving character, base, and tone error channels. Tones are canonical lexical tones from pinyin conversion with the neutral tone as a fifth category; tone sandhi and polyphonic characters are not modeled.

2.2. Panel rule

The candidate inventory is the set of recognizer configurations whose transcripts of these corpora existed before this study. It was assembled in earlier work that inspected speechocean762 correlations. The admission rule below does not use performance, no candidate was added or removed for this study, and the excluded configurations are released with their reasons. Admission is decided before any result is seen: openly downloadable weights; support for the target language or a universal phone inventory; one configuration per acoustic checkpoint, so decoder variants of an admitted checkpoint are excluded; no artificially degraded inputs; no configuration whose checkpoint identity is undocumented. Every configuration that passes is admitted, and none is removed for performance. Applied to the available English transcripts this admits 28 listeners in 10 families (Table 1); the Mandarin panel has 8.

Table 1. English panel by family with mean phoneme error on the speechocean762 training split. Empty outputs (moonshine, 7.5%) score as full deletion; checkpoint identifiers, decoder settings, and transcript hashes are in the released inventory.

GMM-HMM: sphinx-enus (0.65). **Kaldi hybrid:** vosk-small (0.33). **Convolutional CTC:** jasper (0.26), quartznet (0.26). **Self-supervised CTC, LibriSpeech:** wav2vec2-base (0.31), w2v2-large-lv60 (0.23), wav2vec2-robust (0.23), hubert-large (0.25), wavlm-large (0.30). **Self-supervised CTC, other domains:** w2v2-xlsr-cv (0.23), xlsr-1b-cv (0.25), w2v2-swbd (0.31), atc-w2v2 (0.35), atc-xlsr (0.56). **Whisper:** tiny (0.27), tiny.en (0.26), base.en (0.21), small.en (0.17), distil-small.en (0.18), distil-large-v3 (0.14), crisperwhisper (0.12). **Other encoder-decoders:** canary-1b (0.16), seamless-m4t-v2 (0.24), moonshine-base (0.27). **Transducer:** parakeet (0.15). **Speech LLM:** qwen2audio (0.13). **Phone recognizers:** espeak-phone (0.38), allosaurus-eng (0.70).

2.3. Hypotheses and protocol

Seven hypotheses were written before analysis; five (H1, H2, H4, H5, H6) form the primary confirmatory tests, one per

corpus, and H3 and H7 are pre-specified secondary composition analyses. H1: the readout correlates with pronunciation-quality ratings. H2: the panel exceeds the best single listener, chosen on development data. H3: cross-family panels exceed single-family panels of equal size and matched strength. H4: the readout associates more with segmental than with prosodic ratings. H5: it correlates with human word recognition. H6: the tone channel tracks tone ratings more than consonant ratings. H7: high-error listeners alone retain signal.

The speechocean762 training split (2,500 utterances, 125 speakers) was the only development data: readout, panel rule, statistics, and every comparator were fixed there. Confirmatory data were then unlocked once each in a fixed order, speechocean762 test, ERJ, EpaDB, ALLSSTAR, OMPAL, each with one pre-specified primary comparison and a decision rule, after the analysis plan was committed to version control. This guarantees that no confirmatory analysis was inspected before protocol commitment; it does not imply first exposure to these public corpora, which we had used before.² Correlations are Pearson r with 2,000 speaker-clustered (talker-clustered for ALLSSTAR) percentile bootstrap resamples, seed 20260913; paired differences use the same resamples. Secondary analyses are exploratory and uncorrected. The comparators fixed on the training split are the single listener hubert-large, the 9 highest-error listeners (weak third), and 200 cross-family draws matched to the 7 Whisper listeners in mean error (± 0.03) and mean single-listener correlation (± 0.02).

3. DATA

speechocean762 [3] has 2,500 training and 2,500 test utterances from disjoint Mandarin-L1 speakers with five-rater consensus ratings of accuracy, completeness, fluency, prosody, and total. ERJ [14] has 3,800 sentences from 190 Japanese-L1 speakers, rated for segmental quality, rhythm, and intonation on disjoint utterances of the same speakers. EpaDB [15] has 3,160 utterances from 50 Argentinian-Spanish-L1 speakers with a holistic pronunciation-quality phrase score (annotator 1); segmental specificity is tested on ERJ, not here. ALLSSTAR [16, 17] supplies 1,500 clean HINT sentences from 100 talkers of 21 first languages; the criterion is each talker’s mean word-recognition accuracy by L1-English listeners across quiet and noise conditions. OMPAL [18] has 1,768 Mandarin utterances from 46 French-L1 learners with word-level tone and consonant ratings; its transcripts come from a re-decode that kept the text. All transcripts predate this study; files, listener identities, row counts, and empty-output counts are catalogued in the released inventory.

²Analysis-plan commit 0525275 (2026-09-13 20:44 JST) in the released repository; the confirmatory results were committed afterwards. The plan was not lodged with an external registry.

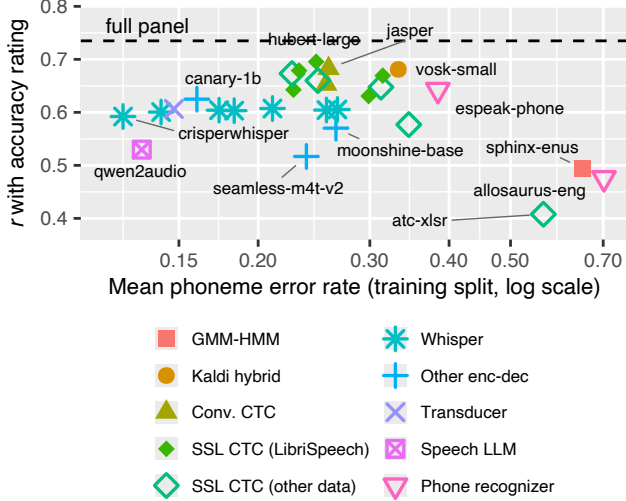


Fig. 1. Each admitted listener on the speechocean762 training split: mean phoneme error rate (log scale) against correlation with the accuracy rating, by family; selected listeners are labeled (all are named in Table 1). The dashed line is the equal-weight PAL panel.

4. RESULTS

4.1. Development split

On the training split the panel correlates with accuracy at $r = 0.735$ [0.678, 0.777] and with completeness at only 0.104; the best single listener, **hubert-large**, reaches 0.695. Random subsets of one, four, eight, and all 28 listeners average 0.604, 0.701, 0.721, and 0.735. The weak third alone reaches 0.714, above the strong third (0.689), and matched cross-family panels of seven exceed the seven Whisper listeners by +0.045 [+0.031, +0.060]. Figure 1 places every listener by error rate and rating correlation: recognition accuracy does not order the listeners as measurement probes, and the panel mean sits above all of them. These observations fixed the comparators; nothing else was changed.

4.2. Pre-specified confirmations

Table 2 lists the five primary comparisons. On the speechocean762 test split PAL reaches accuracy $r = 0.671$ [0.602, 0.725] against 0.633 for the fixed single listener, and 0.648, 0.663, 0.690, and 0.200 for fluency, prosody, total, and completeness. Table 3 places this beside published systems: supervised predictors trained on the benchmark’s ratings score higher, whereas PAL, which uses no ratings, is comparable to published rating-free baselines and to inter-rater agreement. On ERJ the panel correlates with segmental quality at 0.597 [0.558, 0.632] but with rhythm at 0.157 and intonation at 0.073; the segmental-minus-intonation difference is +0.524 [+0.451, +0.597]. On EpaDB the panel

Table 2. Pre-specified primary comparisons, each run once after protocol commitment (SO762 = speechocean762); 2,000-resample clustered 95% intervals. All five decision rules (interval excludes zero) are met.

Data	Primary comparison	Result
SO762 test	H2: PAL — best single, accuracy	+0.038 [+0.021, +0.056]
ERJ	H4: segmental — rhythm	+0.440 [+0.366, +0.516]
EpaDB	H1: r , phrase score	0.415 [0.301, 0.491]
ALLSSTAR	H5: talker r , word recovery	0.576 [0.404, 0.713]
OMPAL	H6: tone — consonant	+0.196 [+0.099, +0.293]

Table 3. Accuracy correlation on the speechocean762 test split. Published values are quoted, not reproduced; human agreement uses the released five-rater scores.

System	Acc. r
<i>Supervised on speechocean762 ratings (published)</i>	
GOPT [1]	0.714
HMamba [2]	0.807
<i>Without expert ratings (published)</i>	
Transcript-guided DTW distance [19]	0.633
TextPA, Gemini-2.0-flash [20]	0.532
<i>Without expert ratings (this work, pre-specified)</i>	
Best single listener, hubert-large	0.633
PAL, equal-weight mean of 28	0.671
<i>Human raters</i>	
Mean pairwise agreement (five raters)	0.618
Each rater versus mean of the other four	0.732

exceeds **hubert-large** by +0.114 [+0.077, +0.152] and reaches 0.695 at the speaker level.

ALLSSTAR replaces expert ratings with the words that human listeners actually recovered; each talker’s 15 utterance scores are averaged before correlation (Fig. 2). The panel exceeds the fixed single listener (0.476) by +0.100 [+0.001, +0.200]; its margin over the seven Whisper listeners (0.548) is +0.028 [−0.001, +0.058]. On OMPAL the tone channel correlates with tone ratings at 0.389 [0.308, 0.459] and with consonant ratings at 0.192, whereas the syllable-base channel gives 0.257 and 0.270. Counting tone errors only on syllables whose base pinyin matches the reference raises the tone correlation to 0.475 [0.402, 0.532] with consonant at 0.120, so the tone signal is not a by-product of base-syllable errors.

4.3. Composition and aggregation

Table 4 answers whether the instrument is more than an average of many recognizers. Every matched cross-family draw scores above the Whisper panel on the test split (100% of draws), but on ALLSSTAR the margin over Whisper includes zero, so family breadth is supported on the development

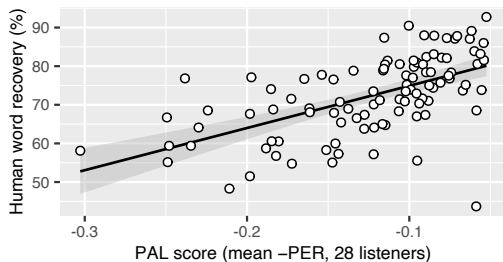


Fig. 2. Panel score and human word recognition for 100 ALLSTAR talkers ($r = 0.576$ [0.404, 0.713]). Each point is a talker; the line is the least-squares fit.

Table 4. Secondary analyses on the speechocean762 test split (accuracy); Δ is paired against the comparator named in the row. Comparators were fixed on the training split.

Configuration	r	Δ [interval]
PAL, equal-weight mean	0.671	—
Weak third (9) vs. hubert-large	0.649	+0.016 [−0.001, +0.034]
Whisper panel (7)	0.621	—
Matched cross-family (200 draws) vs. Whisper	0.649	+0.028 [+0.013, +0.042]
Median over listeners vs. PAL	0.652	−0.019 [−0.026, −0.012]
Minimum error per utterance vs. PAL	0.581	−0.089 [−0.154, −0.032]
Inverse training-error weights vs. PAL	0.666	−0.005 [−0.011, +0.001]

benchmark and not established on the external criterion. The weak third alone reaches 0.649 [0.588, 0.700], close to the fixed best single listener (0.633; paired difference +0.016, an interval that includes zero), the pre-specified secondary comparison for H7; no equivalence margin was pre-specified, so this is a similar correlation, not demonstrated equivalence. Among aggregation rules over all 28 listeners, the median, the per-utterance minimum error, and inverse-training-error weights do not improve on the equal-weight mean. Random subsets of one, four, eight, and 28 listeners average 0.542, 0.637, 0.656, and 0.671, matching the development curve. Two non-pre-specified checks: dropping the two unmapped phone recognizers changes the test and ALLSTAR correlations by −0.001 [−0.003, +0.001] and −0.003 [−0.012, +0.006]; equal weight per family instead of per listener changes them by +0.000 [−0.005, +0.005] and −0.006 [−0.025, +0.014].

4.4. What the panel measures

Three properties follow from the confirmations. First, the readout is mainly segmental: it tracks accuracy, segmental ERJ ratings, and word recovery, but rhythm and intonation only weakly, and its completeness correlation is low. Con-

trolling for accuracy (a non-pre-specified check) reduces its test-split correlations with fluency, prosody, and total from 0.648, 0.663, and 0.690 to partial correlations of 0.272 [0.195, 0.347], 0.312, and 0.233, so most, but not all, of those associations run through shared segmental variance. Second, measurement signal is not recognition quality: the weakest third of the panel achieves a correlation similar to the best single recognizer, while the most accurate recognizers, which recover accented speech almost regardless of pronunciation, contribute little variance. Third, aggregation matters more than curation: eight arbitrary listeners land within a few hundredths of the full panel, equal weights are not improved by medians or error-based weights, and family breadth helped on the development split but not on the external criterion. Random eight-listener subsets already captured most of the full-panel signal here, which bounds the inference cost of a practical panel; whether such smaller panels transfer equally was not tested.

5. LIMITATIONS AND CONCLUSION

The readout is relative: mapping it to a grade requires a labeled calibration set. It has no time variable, and most of its association with fluency and holistic ratings runs through segmental accuracy; a timing channel would be a separate instrument with its own validation. Every reported statistic is reproduced exactly from the released per-listener error rates, the corpora’s published ratings, and the scripts; the transcripts were decoded before model revisions were pinned, so the inventory records reconstructed repository revisions where available and a fresh decode is not guaranteed to be bit-identical. Phone recognizers are compared without inventory mapping. Within these limits, PAL, an equal-weight panel of 28 recognizers admitted by rule, measures prompt-conditioned recoverability without rating labels: it beats the best single recognizer on a held-out split, is segmental rather than prosodic, transfers to a new L1, tracks human word recovery, and extends to transcript-derived lexical-tone recoverability, with each claim tested once under a pre-specified, committed protocol.

6. COMPLIANCE WITH ETHICAL STANDARDS

This study reanalyses existing corpora obtained through their providers’ procedures; no participants were recruited and no new recordings or ratings were collected.

7. ACKNOWLEDGMENTS

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